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AI search

AI search is the process of finding a sequence of actions or steps that lead to from an initial state to a goal state in an Artificial Intelligence system.

Problem solving, game playing, route planning and optimization are some applications of AI search.

ex: AI search algorithms.

- Breadth - First search (BFS)
- Depth - First Search (DFS)

Why is search important in AI ?

- Enabling machines to make intelligent decisions.
- Solve complex problems.
- Navigate through large, complex search spaces to find optimal or near optimal solutions.

AI is for problem solving and for that we build problem solving AI.
Ex. for problem solving agents are,
khan academy, poised.com and sembly.ai

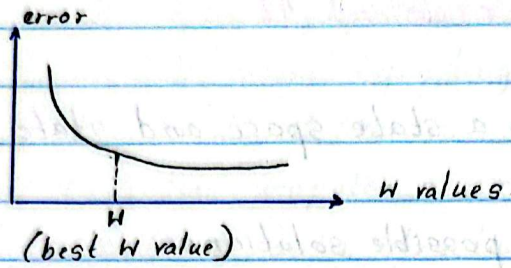
These problem solving agents should do some search.
Three objectives of search are

1- Finding a solution
2- Finding all the solutions
3- Finding the best solution to the problem

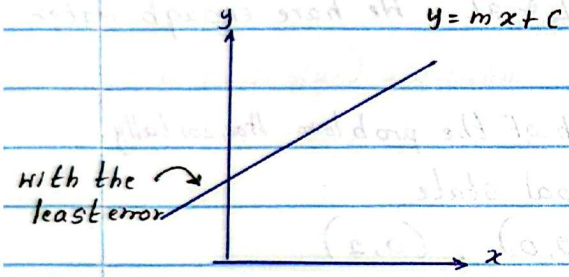
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ex:

01) Finding the best H(Weight) value in a neural network is also a search. Doing multiple training cycles reduces the error of the output in a neural network.
Neural network has many W (weight values)

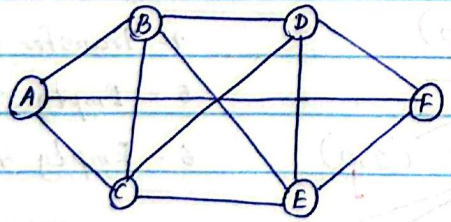


02) Linear Regression (Machine Learning)



We can have number of (m, c) combinations.
We get the best (m, c) by doing a search.

03) In path finding agents.
Objective is to find the shortest distance between A and E.



Here also we do a search and get the shortest out of the multiple possible ways.

eg: Travelling salesman problem.
A, B, ..., F are cities. There is a salesman he should travel each city only once and come back to the same location.
All the cities are connected & paths are bidirectional.
This is an np hard, when problem is larger it cannot be solved.
* This is not solved yet.
n cities → (n-1)! possible results

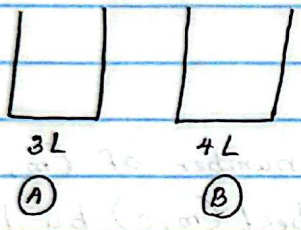
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04) Games also need search to get the path avoiding obstacles and to decide the best moves of the characters. So searching is an essential part in AI.

State Space Search - Core behind AI

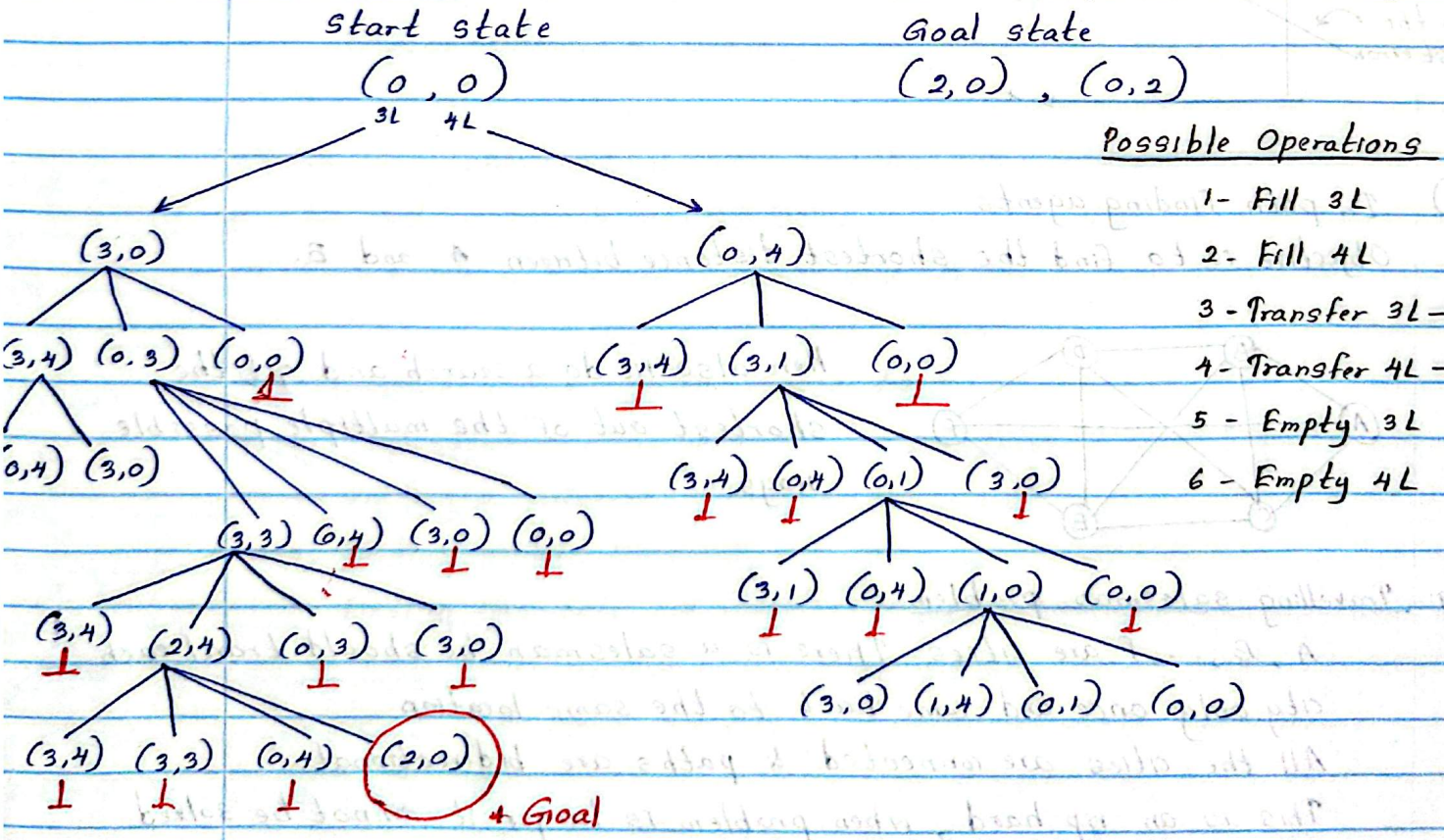
Looking for a state in a state space and state is the solution for the problem.

State space - All possible solutions.



Both (A) and (B) are empty.
Collect 2l of water for A or B where capacities are 4L & 3L. We have enough water.

BFS (Breadth-First Search) - Search for the whole breadth of the problem. Horizontally.



Possible Operations

1- Fill 3L
2- Fill 4L
3 - Transfer 3L → 4L
4 - Transfer 4L → 3L
5 - Empty 3L
6 - Empty 4L

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This method also called,
Blind Search / Exhaustive Search / Uninformed search / Brute force Search

-extra-

- Uniform Cost Search - a search algorithm that finds the lowest cost path by always expanding the node with the smallest total path cost.
- Abdul Bari - An Indian computer science educator known for explaning algorithms, data structures and other CS topics.
- KnapSack problem - An optimization problem where the goal is to select items with the maximum total value without exeeding a given weight limit.
- A* (A-star) algorithm - An informed search algorithm that finds the shortest path by using a formula.

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